ONE-WAVE FRAMEWORK

Book 2 — Small

Chapter 1: The Cell — Hexagonal Compression Unit

Note: relocated from Book 1 Chapter 3. Content unchanged from the original — a biological cell is far larger than atomic/subatomic scale, so this chapter belongs at the Small tier, not Micro.

Version: 4.0 Date: July 22, 2026 Class: B — Applied Layer Spine: Gray / 2D / 3D / Mathematics / Predictions / Yellow Audit / Future Work / Closing Thoughts

Dependencies: B-06 Paired Loop, B-06b Four Views, E-03 Surface, E-06 Scale-Invariant Loop, B-08 Threshold Windows, A-10 Persistent Mode Status: GREEN (structure) / YELLOW (formal mapping)

Gray — Standard Model Reference

In conventional biology, a cell is a membrane-bound unit containing: nucleus (DNA), organelles (mitochondria, ribosomes, etc.), cytoplasm, and a lipid bilayer membrane.

Cells communicate via: - Chemical signaling (neurotransmitters, hormones) - Electrical gradients (ion channels, action potentials) - Gap junctions (direct cytoplasmic connections)

The nervous system transmits information via action potentials — electrical spikes of approximately 100mV that travel along myelinated axons at speeds of 1-120 m/s depending on fiber diameter and myelination.

Signal integration: neurons sum inputs at the cell body (soma). If the sum exceeds the firing threshold (~-55mV), an action potential fires.

Standard Model strength: extremely detailed mechanistic understanding of ion channels, neurotransmitters, receptor binding, signal cascades. Standard Model limitation: no unified framework connecting cellular dynamics to the same physics governing galaxies, protons, or consciousness.

2D One-Wave Interpretation

In 2D, a cell is a hexagonal compression unit.

The hexagon is the most efficient tiling of 2D space: six equal sides, minimum perimeter for a given enclosed area, minimum surface energy (E-03) for maximum internal volume.

A signal enters one face of the hexagon. It is processed through the internal oscillation (Paired Loop, B-06). It is expressed out the opposite face.

The cell is a single paired-loop node (B-06): one compression and one expression per cycle. Express -> Compress -> Threshold -> Return.

A proposed cross-scale extension groups local units into larger sensor and neural hierarchies. That extension is not required to define the biological cell itself. Its consciousness interpretation is now developed in Books/Proposed_One_Wave_Consciousness, and its build form is developed in Books/Proposed_Android_Brain and C-312.

3D One-Wave Interpretation

In 3D, cells are the micro-scale expression of the same lattice dynamics governing everything else in the framework.

The hexagonal geometry gives minimum surface energy (E-03) for a given internal volume. This is why biological cells tend toward spherical and hexagonal geometries — not by design, but because minimum-energy surfaces are attractors in the lattice.

The cell membrane is the surface tension layer (E-03) holding the internal excitation in. It is the boundary of the Persistent Mode that is the cell.

Cross-scale neural and consciousness hypotheses were previously embedded here. They are preserved in History/Pre_Reorganization/Book2_Ch01_The_Cell_v3_original.md and actively developed in Books/Proposed_One_Wave_Consciousness. Their engineering implementation is developed in Books/Proposed_Android_Brain.

This chapter now limits its canonical biological claim to the cell as a bounded signaling and persistent-mode system. It does not close the brain/consciousness hypothesis; it gives that hypothesis the correct active research home and status.

Mathematics

Hexagonal packing — minimum perimeter P for area A: P_hex = sqrt(4*pi*A / (3*sqrt(3)/2))

Paired loop at cellular scale (B-06): Express -> Compress -> Threshold (action potential) -> Return or Break (fire or not)

Threshold at cellular level is proposed to map to B-208 Threshold Windows, but the numerical windows are under active reconciliation and must not be treated as biologically calibrated values.

Signal propagation speed: v_signal = Delta_x / Delta_t (one lattice update = one flip)

Predictions

1. Cellular signal propagation obeys the same threshold structure as B-08 Threshold Windows. Prediction: firing threshold, adaptation, and fatigue all correspond to specific threshold bands.
2. Hierarchical neural compression may show recurring ratios, including a proposed 3:1 candidate, but this is now tracked as an active cross-scale hypothesis rather than a settled cellular prediction.
3. Action potential is a Persistent Mode (A-10) traveling along a 1D lattice chain. Prediction: action potential propagation speed is proportional to sqrt(beta).
4. Hexagonal cell geometry emerges from minimum surface energy (E-03). Prediction: any system tiling 2D space under minimum-energy constraints produces hexagonal geometry.
5. The four views may support a functional comparison with neural coordination. That comparison is active hypothesis work in G-719 and the Proposed One-Wave Consciousness book, not a settled anatomical mapping.

Yellow Audit

- Exact mapping of action potential to Persistent Mode not yet formalized
- Chemical gradient as nabla_psi term needs derivation
- Receptor binding as Paired Loop coupling event needs formalization
- Proposed hierarchical compression ratios remain uncalibrated
- Exact mapping of biological thresholds to B-08 windows not yet done
- Four-views neural mapping remains an active functional hypothesis in G-719

Future Work

Simulate hexagonal cell network signal propagation. Map cellular threshold bands to B-08 windows with biological data. Formalize action potential as Persistent Mode propagation. Continue the active M4 and consciousness hypothesis in the proposed consciousness book.

Closing Thoughts

The chemical gradient is the nabla_psi term (A-04 Gradient). The action potential is a Persistent Mode traveling along a 1D lattice chain. Receptor binding is a Paired Loop coupling event (B-06). The nervous system may be a biological scale instance of the same recursive architecture; this remains an active hypothesis requiring explicit mapping and tests.

Nothing new was imported. Same update rule, same paired loop, same threshold windows. The biology is the framework at micro scale. The galaxy is the framework at macro scale. Same algorithm. Different participants. Different frequencies.

END OF BOOK 2 CHAPTER 1 One wave. Mirror builds. Mark Wright. Kitty Hawk V0.